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DS 4200
Visualizations Design Explanations and Takeaways

Published Website Link: <https://gokulramanan920.github.io/Life-Expectancy-Factors/>

Bubble Chart (Wealth & Health of Nations) → Altair

Description:

The x-axis represents the GDP per capita (scaled to US dollars) of a nation. The y-axis represents the average life expectancy of a nation. The size of a data point is indicative of the population of a nation, allowing us to account for the population alongside the GDP per capita. The data is also separated by region to understand trends between regions and globally. Additionally, an animation between the countries and their GDP per capita over the years is used to display and interpret how changes in GDP affects the life expectancy within a country itself. The purpose of the visualization is to classify any trends or patterns between income (measure of gdp per capita) and life expectancies.

Findings:

The animated bubble chart demonstrates a clear positive correlation between GDP per capita and life expectancy, with a notable threshold effect shown by the logarithmic scale. At lower income levels, the relationship is at its most volatile as moving from extreme poverty (like Burundi at \$208 GDP per capita with 48 years life expectancy in 1990) to lower-middle income produces extremely steep gains in longevity. However, the steepest improvements occur between roughly \$500 and \$10,000 per capita. Beyond approximately \$15,000 - 20,000 per capita (such as Canada at \$21,000 with 77.3 years in 1990), additional wealth yields slightly diminishing returns on life expectancy. China's trajectory exemplifies this pattern where it starts at \$318 GDP per capita with 70 years life expectancy in 1990 (notably high for its income level), it reaches \$10,262 and 77 years by 2019, demonstrating both the strong correlation and the tapering effect as countries develop. The animation reveals that virtually all countries moved upward and rightward over the three decades, with African nations showing improving trajectories from the lower-left cluster while European nations in the upper-right showed more modest gains, having already achieved high life expectancies. While medical and healthcare advancements certainly contributed to global life expectancy improvements from 1990 - 2019, the persistent clustering by income level and the divergent trajectories between wealthy and poor nations underscore GDP per capita as a fundamental driver of population health outcomes, likely through its enabling of healthcare infrastructure, nutrition, sanitation, and other life-extending resources.

Barplot Dual-Axis Visualization (Gov Expenditure vs Life Expectancies) → Altair

Description:

This visualization combines a bar chart with an overlaid scatter plot to compare government expenditure patterns and life expectancy across continents. The colored bars represent the average life expectancy for each continent (left y-axis in years), while the black circular points overlaid on each bar show the average government expenditure as a percentage of budget (right y-axis). The expenditure type can be changed using the dropdown menu to explore health, education, or military spending. Each continent is represented by a distinct color for readability and quick regional comparisons. By aggregating data at the continental level, this chart reveals broader patterns in how government spending priorities correlate with population health outcomes, making it easier to identify which regions invest more heavily in specific sectors and how those investment patterns relate to life expectancy across the globe.

Findings:

Our analysis reveals distinct patterns across different expenditure categories. For health spending, there is a strong positive correlation with life expectancy. Europe leads with both the highest average life expectancy (77 years) and the highest health expenditure (5.8%), followed closely by Oceania and North America. Africa shows the lowest life expectancy (58 years) with minimal health spending (2.6%), demonstrating the critical impact of healthcare investment on population health outcomes. For education spending, the relationship is less clear. While Europe, North America, and Oceania maintain high life expectancies, their education expenditure varies significantly (ranging from 12% to 17%). Africa shows the highest education spending (17%) yet has the lowest life expectancy, suggesting that education expenditure alone does not directly drive life expectancy improvements without complementary healthcare infrastructure. For military spending, there appears to be an inverse relationship. Asia shows the highest military expenditure (12%) with moderate life expectancy (71 years), while Europe has the lowest military spending (4%) yet the highest life expectancy. North America maintains relatively low military expenditure (2.5%) with high life expectancy (74 years). This pattern suggests that nations prioritizing military spending over other sectors may see diminished returns on population health. Overall, health expenditure emerges as the only government spending category with a clear, positive correlation to life expectancy. The continental-level aggregation reveals that regions investing more in healthcare infrastructure consistently achieve better health outcomes, while military and education spending show no consistent relationship with longevity.

KDE Plot (Sanitation vs. Life Expectancy) → D3

Description:

This density plot shows how safely managed sanitation relates to life expectancy across different countries and years. Each point represents a country-year observation, and the shaded density in the background helps show where most of the points are concentrated. Darker areas indicate higher concentrations of observations. Using both the scatter points and the density layer makes it easier to see overall patterns compared to a basic scatterplot. The goal of this figure is to highlight the general relationship between sanitation access and longevity. Since there are many overlapping data points, the density shading helps reveal clusters and trends that might be hard to see otherwise.

Findings:

The plot shows a clear positive relationship between sanitation access and life expectancy. Countries with very low sanitation coverage, usually below about 20 percent, display a wide range of life expectancies. Some fall below 50 years, while others reach into the mid-70s. This spread suggests that poor sanitation does not affect all countries in the same way. Other factors, such as healthcare systems, income levels, and environmental conditions, also play important roles in shaping outcomes at low sanitation levels.

As sanitation coverage increases into the middle range, roughly 30 to 60 percent, life expectancy rises more consistently into the 70 to 80 year range. The density of points becomes tighter here, which suggests that expanding access to safe sanitation is strongly associated with better public health outcomes.

At higher sanitation levels, usually above 70 to 80 percent, the observations cluster mainly in the upper-right part of the plot. Countries with widespread access to safely managed sanitation typically reach life expectancies between 75 and 85 years, with much less variation. This suggests that once sanitation coverage becomes widespread, countries tend to maintain strong and stable health outcomes.

Overall, the plot shows that improvements in sanitation are closely linked to longer life expectancy. The biggest gains appear as countries move from low to moderate sanitation coverage, while countries with near-universal sanitation experience the highest and most consistent life expectancy levels.

Boxplot Visualization (Healthcare Professionals vs Life Expectancy) → Altair

Description:

There are three separate box plots. Each boxplot represents a third of the life expectancies (sorted). As a result, the first box plot represents the lowest third of life expectancies globally. The second box plot represents the middle third, and the last box plot represents the upper third. Each boxplot showcases the spread of a healthcare staff member among the countries in those groups of life expectancies. The purpose of the data is to showcase the spread of healthcare staff

for countries with lower life expectancies in contrast to those with higher life expectancies (on average). Healthcare staff members include doctors, pharmacists, and nurses & midwives. The visualization allows us to understand whether there is a correlation to healthcare staffing and life expectancies.

Findings:

We found that generally for all three healthcare staff (doctors, nurses & midwives, pharmacists), that countries with higher life expectancies have more staffing. For example, the lower third countries (in terms of life expectancy) have a median doctor count of 3 per 10,000 with the third quartile hitting 20/10,000. However, the middle third has a greater median and Q3 with 16.4 and 28.9 respectively. The trend continues in the upper third with a median of 29 and Q3 of 35.6. Such a trend indicates that a greater staffing in doctors correlates to greater life expectancies. We can assume that greater numbers of doctors also relate to better staffed hospitals that receive more funding, which allows for more resources to be put into healthcare overall. However, the trend is not well illustrated when comparing the nurses & midwives between the thirds. The lower and upper thirds contrast heavily through having a 49.6 point difference in the median of staffing of such professions, but the middle third indicates a lower median than the lower third. We believe this to be the case considering there was unfortunately not enough data regarding this profession in the countries involved in the middle third of life expectancies. The trend picks back up for pharmacists, indicating a positive correlation of pharmacists to life expectancies.

Scatterplot Visualization (Fruit Diet vs Life Expectancies) → Altair

Description:

The x-values present the amount of calories in fruit & vegetables that a person may consume on average per day in a specific country. The y-values correlate to the average life expectancies of people in a specific country. The data is separated into regions to allow for better understanding of which regions overall follow certain patterns in terms of their dietary habits, allowing us to dissect the data further. Additionally, the visualization allows for zooming in and out for better clarity regarding data points in a specific region or quadrant. Moreover, there's regional selection on the legend as well as a LOWESS trend curve to capture the general trend of the data distribution. The purpose of the plot is to understand whether healthy dietary habits such as eating fruits/vegetables contribute positively to the average life span of individuals globally.

Findings:

The scatterplot reveals a more nuanced relationship than simple positive correlation as it forms an inverted U-shape pattern shown by the LOWESS curve. Life expectancy rises steeply as fruit and vegetable consumption increases from very low levels (such as Chad at 28 kcal/day

with 48 years life expectancy) up to an optimal range of approximately 150-300 kcal per day, where life expectancy peaks around 73-74 years. However, beyond 300 kcal per day, the curve descends, with countries showing very high consumption experiencing declining life expectancy. For instance, while Bermuda consumes 462 kcal/day with 77 years life expectancy, this represents an outlier rather than the trend, as most countries with similarly high consumption show lower life expectancies. This pattern likely reflects confounding variables rather than causation because extremely high fruit and vegetable consumption as a proportion of total diet may indicate limited access to other essential nutrients like proteins and fats, suggesting food insecurity or dietary imbalance. European nations cluster in the optimal range with high life expectancies, while African countries predominantly occupy the lower end of consumption with correspondingly lower life expectancies. Within Africa itself, countries with higher fruit and vegetable intake show improved life expectancy, following the upward portion of the curve before the optimal threshold.